

■ PLANT LIVING DATABASE

# Deployment Guide

Set up the Plant Living Database PWA with Google Sheets sync and GitHub Pages hosting.

5

Setup  
Parts

30-45

Minutes  
First Time

Free

Hosting  
Cost

PWA

Any  
Device

Overview

The Plant Living Database is a Progressive Web App (PWA) that runs entirely in the browser. Plant data lives in browser `localStorage` and is optionally backed up to a private Google Sheet via a small Apps Script web service. The app is hosted for free on GitHub Pages.

What You Will Need

|  |                                                                                                                                                                                     |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | A Google account (free Gmail is fine)                                                                                                                                               |
|  | A GitHub account (free at <a href="https://github.com">github.com</a> )                                                                                                             |
|  | The files from PD.zip: <code>index.html</code> , <code>plantdatabase.html</code> , <code>manifest.json</code> , <code>sw.js</code> , <code>icons/</code> , and <code>Code.gs</code> |
|  | About 30–45 minutes for initial setup                                                                                                                                               |

## PART 1

**Set Up Google Sheets & Apps Script Backend**

The Apps Script acts as a web server that reads and writes your plant data to a private Google Sheet in your own Google Drive.

1

**Create a new Google Sheet**

Go to [sheets.google.com](https://sheets.google.com) and click + Blank. Name it PlantDB.

**Open the Apps Script editor**

In your spreadsheet click Extensions → Apps Script. A new tab opens with the script editor.

3

**Paste the Code.gs backend**

Select all code in the editor (Ctrl+A), delete it. Open Code.gs from PD.zip, copy its entire contents, paste into the editor, and click the ■ Save icon.

**Run the setupSheet function**

In the toolbar select setupSheet from the function dropdown, then click ■ Run. Accept the permissions prompt → Review Permissions → choose your account → Allow. You will see "Execution completed" in the log. This creates the Plants and Meta sheets.

5

**Deploy as a Web App**

Click Deploy → New deployment. Set type to Web App. "Execute as": Me. "Who has access": Anyone. Click Deploy, then Authorise access if prompted. Copy the Web App URL — it ends in /exec. You will need this in Part 3.

**■ Re-deploying after code changes**

Every time you modify Code.gs you must create a NEW deployment (Deploy → New deployment). The URL changes each time — update it in the app's ■ Sync page whenever this happens.

## PART 2

## Host the App on GitHub Pages

GitHub Pages is a free static web-hosting service. Your app files live in a GitHub repository and are served at a permanent public URL.

1

**Create a GitHub account**

Go to [github.com](https://github.com) and sign up for a free account. Your username will appear in the app URL.

**Create a new repository**

Click + New repository. Name it plantdb (or anything you prefer). Set it to Public. Tick "Add a README file". Click Create repository.

3

**Upload the app files**

Click Add file → Upload files. Drag and drop from PD.zip: index.html, plantdatabase.html, manifest.json, sw.js, and the icons/ folder. Add a commit message such as "Initial upload" and click Commit changes.

**Enable GitHub Pages**

Go to your repository's Settings tab → Pages (left sidebar). Under Source, select Deploy from a branch. Choose branch: main and folder: / (root). Click Save. After 1–2 minutes your app is live at <https://YOUR-USERNAME.github.io/plantdb/>

## Alternative Hosting Options

| Service                 | How to deploy                                                                                                           | Custom domain |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Netlify Drop</b>     | Drag your project folder onto <a href="https://app.netlify.com/drop">app.netlify.com/drop</a> . Live in under a minute. | Free plan     |
| <b>Vercel</b>           | Import GitHub repo at <a href="https://vercel.com">vercel.com</a> — auto-deploys on every push.                         | Free plan     |
| <b>Cloudflare Pages</b> | Connect GitHub repo at <a href="https://pages.cloudflare.com">pages.cloudflare.com</a> . Global CDN, fast worldwide.    | Free plan     |

## PART 3

**Connect the App to Google Sheets (■ Sync)**

This step links the hosted app to your Apps Script backend. Once connected, your plants are accessible and synced across all your devices.

**Open the live app**

- 1 Navigate to your GitHub Pages URL (e.g. <https://yourusername.github.io/plantdb/>). The app loads with any existing plant data.

**Open the Sync tab**

Click the ■ Sync tab in the navigation bar at the top of the page.

**Paste your Apps Script URL**

- 3 In the "Apps Script Web App URL" field, paste the /exec URL copied at the end of Part 1. If you set an auth token in Script Properties, enter it in the Auth Token field.

**Connect & Test**

Click Connect & Test. A green "✓ Connected" badge and log entry confirm success. If it fails, see the Troubleshooting page.

**Push your data to the cloud**

- 5 Click Push to Cloud. All local plants are uploaded to the Google Sheet. The progress bar and log show upload status.

**Enable auto-sync (optional)**

Tick "Push on every save" under Auto-Sync Settings to automatically push after every add, edit, or delete. Tick "Pull on startup" to always get the latest data when the app loads on any device.

**■ Common causes of connection failure**

(1) You forgot to click Deploy after pasting code. (2) "Who has access" was not set to Anyone. (3) The URL was copied incompletely — it must end in /exec. (4) A new deployment was created but the old URL is still in the app — update it in ■ Sync.

## PART 4

## Install on Your Devices

Once hosted, the app can be installed on any device like a native app — its own icon, full-screen mode, and full offline support.

| ■ PC / Mac (Chrome or Edge)                                                                                                                                                                 | ■ Android (Chrome)                                                                                                                            | ■ iPhone / iPad (Safari)                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Look for the install icon in the address bar (computer with down arrow). Click it and click Install. The app opens in its own window and appears in your Start Menu or Applications folder. | Tap the three-dot menu → "Add to Home screen" or "Install app" → tap Add. The PlantDB icon appears on your home screen and opens full-screen. | Open the URL in Safari (not Chrome). Tap the Share button (↑) at the bottom → scroll down → tap "Add to Home Screen" → tap Add. PWA install on iOS only works in Safari. |

## PART 5

## Day-to-Day Use, Multi-Device Sync &amp; Updates

|                          |                                                                                                                                                          |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Auto-push on save</b> | Every save or delete triggers a push to the Google Sheet automatically if "Push on every save" is enabled in ■ Sync → Auto-Sync Settings.                |
| <b>Offline use</b>       | Changes queue while offline and sync automatically when you reconnect (if "Offline queue" is enabled). A yellow badge shows how many changes are queued. |
| <b>Pull from cloud</b>   | To see changes made on another device: ■ Sync → Pull from Cloud.                                                                                         |
| <b>Sync Now</b>          | Drains the upload queue immediately without waiting for a save event.                                                                                    |
| <b>Open Sheet</b>        | Jump directly to the Google Sheet to view or audit data.                                                                                                 |
| <b>Multiple devices</b>  | Each device connects using the same Apps Script URL. Push from one device, then Pull on another to share data.                                           |

## Updating the App (new version released)

## 1 Open your GitHub repository

Navigate to github.com and find your PlantDB repository.

## Replace the files

Upload the new files from the updated PD.zip using Add file → Upload files. Commit the changes.

## 3 Hard refresh on devices

Ctrl+Shift+R on PC/Mac. On mobile, close and reopen the app. Your plant data is unaffected — it lives in localStorage and Google Sheets.

## Troubleshooting

|                                                    |                                                                                                                                                        |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q: Changes not appearing on another device</b>  | A: Open ■ Sync → Pull from Cloud on that device. Ensure the same Apps Script URL is entered on both devices.                                           |
| <b>Q: "Connected" never appears</b>                | A: Check internet connection. Verify the URL ends in /exec. Confirm the deployment is still active in Google Drive → Apps Script → Manage Deployments. |
| <b>Q: App shows old version after update</b>       | A: Hard refresh: Ctrl+Shift+R (PC), Cmd+Shift+R (Mac). On mobile, clear site cache and reopen.                                                         |
| <b>Q: PWA install prompt does not appear</b>       | A: The prompt shows once per domain. Check the address bar for an install icon. On iOS, always use Safari → Share → Add to Home Screen.                |
| <b>Q: Google Sheet shows wrong or missing data</b> | A: Do not edit the Plants or Meta sheets manually. If corrupted, use ■ Sync → Push to Cloud from the app to restore the sheet.                         |
| <b>Q: Apps Script returns "Unauthorized"</b>       | A: If you set PLANTDB_TOKEN in Script Properties, ensure the same token is entered in ■ Sync → Auth Token field.                                       |
| <b>Q: "Who has access" error on deploy</b>         | A: "Execute as" must be Me and "Who has access" must be Anyone (not Anyone with Google account).                                                       |
| <b>Q: Offline queue not flushing</b>               | A: Ensure "Offline queue" is ticked in ■ Sync → Auto-Sync Settings. Check internet connection, then click Sync Now.                                    |

### ■ Security tip: protect your Apps Script endpoint

By default the /exec URL is public. To restrict access: in Apps Script go to Project Settings → Script Properties → add key PLANTDB\_TOKEN with any value you choose. Then enter the same value in ■ Sync → Auth Token. All requests without the correct token are rejected.